

Interreg



Co-funded by
the European Union

RESPOND

Alpine Space

RESPOND is co-financed by the European Development Fund
through the Interreg Alpine Space programme 2021-2027

EbA strategies **Agroforestry**



About this EbA strategy:

Agroforestry is an integrated agricultural practice that combines traditional farming techniques with forest management, integrating woody and perennial plants with herbaceous crops and/or animal husbandry. This approach aims to create land-use systems that are more productive, healthy, diversified, and sustainable. The integration of trees, shrubs, annual crops, and animals into vineyards is an ancient practice, particularly in the Mediterranean region.

The different types of agroforestry:

- Silvo-arable systems involve growing tree species (woody plants, fruit trees, or others) alongside herbaceous crops.
- Forestry and pastoral systems combine livestock with arboriculture, either for wood or fruit production.
- Cultivations in the forest primarily focus on non-wood products like mushrooms and berries.

Agroforestry emphasises long-term land-use practices and woodlot management, meeting criteria of being intensive, interactive, and integrated. It is a nature-based solution that supports the development of multifunctional landscapes and contributes to enhanced climate resilience.

Key ecosystem services provided:

- Soil health improvement (*enhanced organic matter, nutrient cycling, and soil structure*)
- Erosion control (*root stabilisation*)
- Water retention (*reduced surface runoff*)
- Microclimate regulation (*wind buffering, shading, and temperature moderation*)
- Habitat provision and biodiversity conservation (*refuge, food resources, and ecological connectivity*)
- Natural pest control (*support for beneficial insects and natural enemies*)
- Water regulation (*improved infiltration, retention, and reduced nutrient leaching*)
- Increased overall production and economic diversification (*multiple outputs such as crops, fodder, timber, or non-timber products*)
- Cultural and landscape aesthetic value (*cultural identity, structural diversity and visual character of agricultural landscapes*)

Potential challenges:

- **Management and labour costs:** Agroforestry systems require more complex management, including additional training and careful machinery use, which can increase labour demands and costs compared to monocultures, especially during establishment.
- **Yield uncertainty and economic viability:** The transition from conventional to nature-based agriculture may result in temporary yield reductions, particularly in the early years, while the full land-use and economic potential of by-product-producing species (e.g. poplar or oak) is not always ensured due to local management constraints.
- **Resource competition:** Even though trees can enhance water availability, there is a potential for competition for water and nutrients between trees and the vine, particularly in arid regions or during establishment.